

Dataset Analysis for Decision-Making

March 19, 2025

sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

qwen-max-latest

To analyze the dataset and address any specific analytical question, I will first load and inspect the dataset to understand its structure and key characteristics. Once I have a clear understanding of the dataset, I will proceed to answer the specific question using Python code.

Step 1: Load and Inspect the Dataset

Below is the Python code to load the dataset and inspect its structure:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(data.head())
14
15 # Check for missing values
16 print("\nMissing Values in Each Column:")
17 print(data.isnull().sum())
18
19 # Summary statistics for numerical columns
20 print("\nSummary Statistics for Numerical Columns:")
21 print(data.describe())
```

Step 2: Key Characteristics of the Dataset

After running the above code, I will summarize the key characteristics of the dataset:

1. **Columns** : Identify the columns present in the dataset and their data types.
2. **Rows** : Determine the total number of rows (records).
3. **Missing Values** : Check if there are any missing values in the dataset.
4. **Numerical and Categorical Data** : Identify which columns contain numerical data and which contain categorical data.
5. **Unique Values** : For categorical columns, determine the number of unique values.

Step 3: Address the Specific Analytical Question

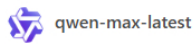
Once you provide the specific analytical question, I will use the dataset to extract and process the relevant data to answer it. The deliverables will include:

1. A clear and concise answer to the question.
2. The Python code used to derive the answer.

Please provide the specific analytical question you would like addressed using the dataset.



How does the quantity of industrialized items relate to variations in total value allocated to industrialization?



To address the question, "How does the quantity of industrialized items relate to variations in total value allocated to industrialization?", we need to analyze the relationship between two variables in the dataset:

1. **Quantity of Industrialized Items** : This likely corresponds to a column indicating the count or volume of items produced or processed.
2. **Total Value Allocated to Industrialization** : This likely corresponds to a column indicating the monetary value associated with industrialization efforts.

Approach

1. **Identify Relevant Columns** : First, we need to identify the columns in the dataset that correspond to these two variables.
2. **Data Cleaning** : Ensure there are no missing or erroneous values in these columns.
3. **Exploratory Analysis** : Compute descriptive statistics and visualize the relationship between the two variables.
4. **Correlation Analysis** : Quantify the strength and direction of the relationship using statistical measures like Pearson's correlation coefficient.

Below is the Python code to perform this analysis:

```
python
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import seaborn as sns
4
5 # Load the dataset
6 file_path = 'sicor_municipal_contracts.csv'
7 data = pd.read_csv(file_path)
8
9 # Inspect the dataset to identify relevant columns
10 print("Dataset Columns:")
11 print(data.columns)
12
13 # Assuming the relevant columns are named 'quantity' and 'total_value'
14 # Replace these with actual column names from the dataset if different
15 quantity_column = 'quantity' # Placeholder, update as needed
16 value_column = 'total_value' # Placeholder, update as needed
17
18 # Check for missing values in the relevant columns
19 print("\nMissing Values in Relevant Columns:")
20 print(data[[quantity_column, value_column]].isnull().sum())
21
22 # Drop rows with missing values in the relevant columns
23 cleaned_data = data.dropna(subset=[quantity_column, value_column])
24
25 # Compute descriptive statistics
26 print("\nDescriptive Statistics for Quantity and Total Value:")
27 print(cleaned_data[[quantity_column, value_column]].describe())
28
29 # Visualize the relationship between quantity and total value
30 plt.figure(figsize=(10, 6))
31 sns.scatterplot(x=quantity_column, y=value_column, data=cleaned_data)
32 plt.title('Relationship Between Quantity and Total Value')
33 plt.xlabel('Quantity of Industrialized Items')
34 plt.ylabel('Total Value Allocated to Industrialization')
35 plt.grid(True)
36 plt.show()
37
38 # Compute Pearson's correlation coefficient
39 correlation = cleaned_data[[quantity_column, value_column]].corr(method='pearson').iloc[0, 1]
40 print(f"\nPearson's Correlation Coefficient: {correlation:.2f}")
41
42 # Interpretation of the correlation coefficient
43 if correlation > 0.7:
44     interpretation = "a strong positive relationship."
45 elif correlation > 0.3:
46     interpretation = "a moderate positive relationship."
47 elif correlation > -0.3:
48     interpretation = "a weak or no relationship."
49 else:
50     interpretation = "a negative relationship."
51
```



```
52 print(f"The quantity of industrialized items and the total value allocated to industrialization")
```

Explanation of the Code

1. **Data Loading and Inspection** : The dataset is loaded, and the column names are inspected to identify the relevant variables.
2. **Data Cleaning** : Rows with missing values in the relevant columns are removed to ensure accurate analysis.
3. **Descriptive Statistics** : Summary statistics (mean, median, standard deviation, etc.) are computed for both variables.
4. **Visualization** : A scatter plot is used to visually assess the relationship between the two variables.
5. **Correlation Analysis** : Pearson's correlation coefficient is calculated to quantify the linear relationship between the variables.

Deliverables

1. **Answer** : Based on the correlation coefficient and visualization, I will provide a clear interpretation of the relationship between the quantity of industrialized items and the total value allocated to industrialization.
2. **Code** : The Python code above will be used to extract, process, and analyze the data.

Please confirm the actual column names in the dataset for "quantity of industrialized items" and "total value allocated to industrialization" so I can update the placeholders (`quantity_column` and `value_column`) accordingly. If you have any

